

Figures Bundle

Decoupling Cyclone-Induced Saline Inundation from Agronomic Flooding in Sentinel-1/2 Rice Phenology Retrieval: A Multi-Year Bay-of-Bengal Coastal Framework (2017-2024)

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All figures are embedded at publication size (6.0-inch column width) from PNGs in figures/ regenerated from analysis/results/real_v21/. Each figure carries a short caption immediately below the image.

Figure 1. figure1_study_area.png

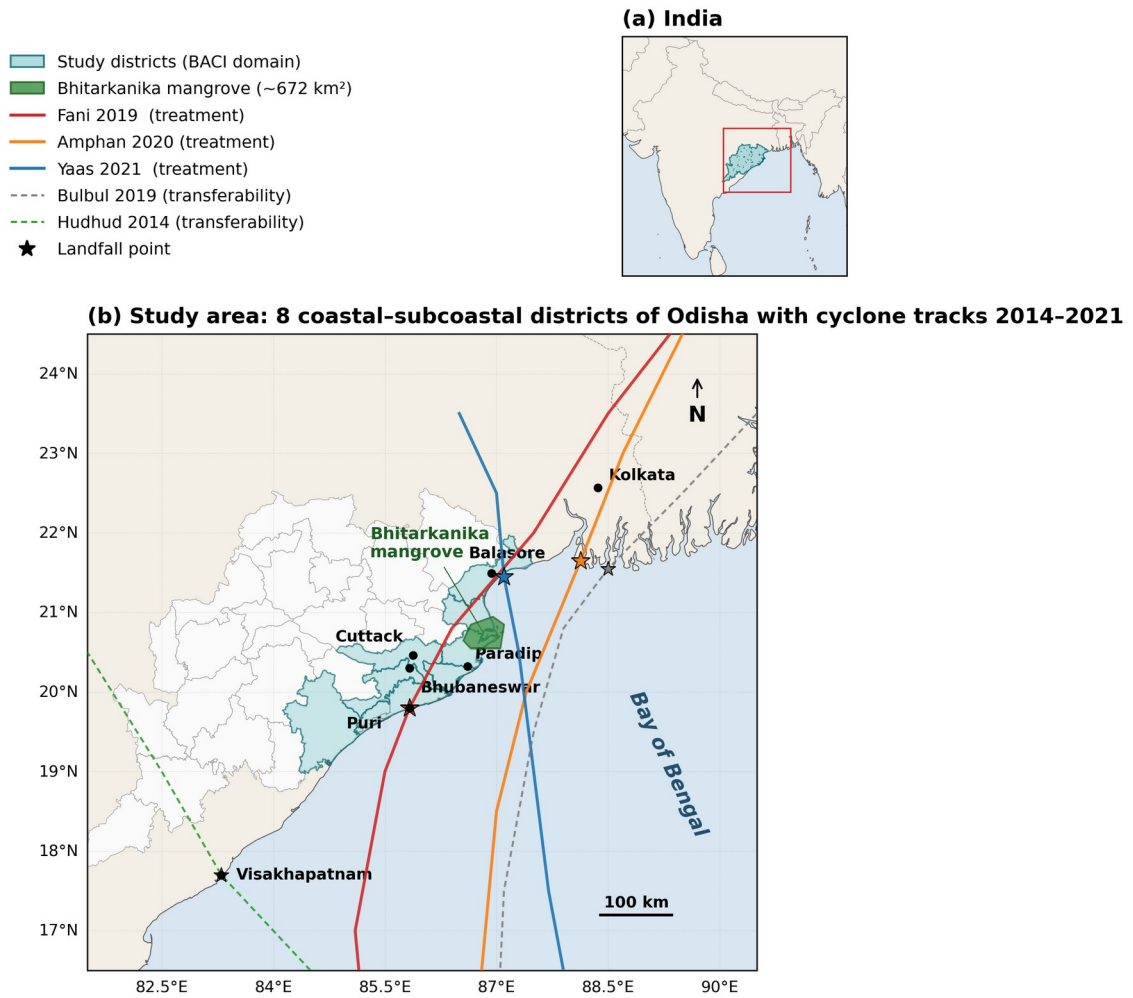
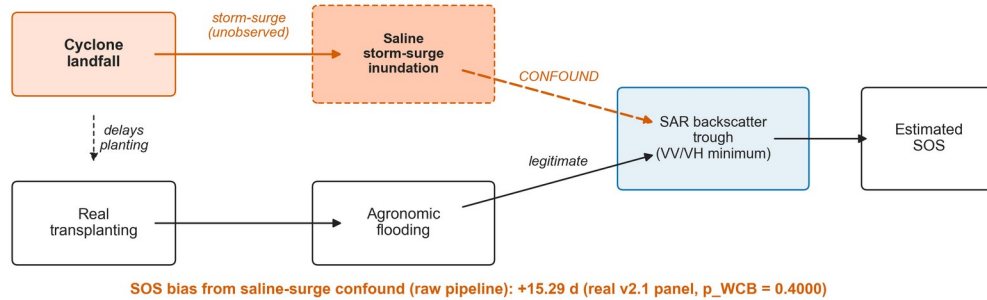


Figure 1. Study area: coastal and inland Odisha BACI districts. Treated coastal districts (Balasore, Bhadrak, Kendrapara, Jagatsinghpur, Puri) and inland control districts (Dhenkanal, Angul, Cuttack), with IBTrACS cyclone tracks for Fani (May 2019), Bulbul (Nov 2019), Amphan (May 2020), and Yaas (May 2021).

Figure 1B. fig1b_identification_dag.png

(a) Naive pipeline (every prior cyclone-affected SAR rice study)

saline storm-surge inundation is mis-read as transplanting flooding, biasing SOS



(b) Corrected pipeline (this study)

the saline-flood classifier (Module 02) breaks the storm-surge → trough arrow, restoring identification

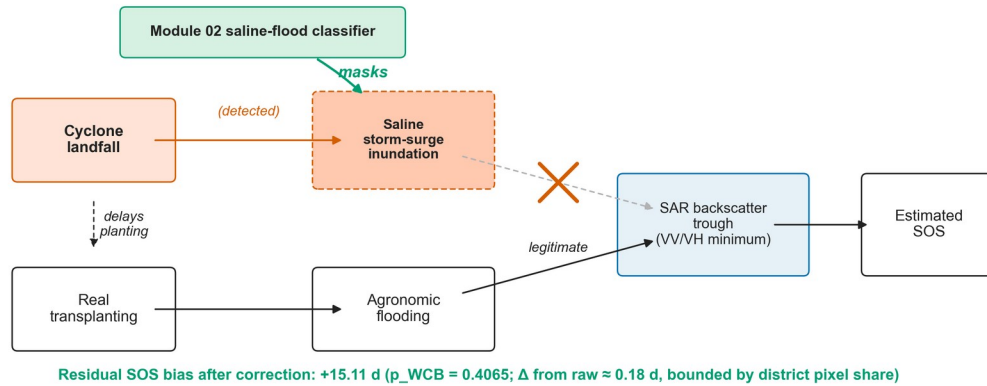


Figure 1B. Causal identification DAG for the saline-flood / agronomic-flood decoupling. The cyclone landfall is a single exogenous shock that opens parallel legitimate (transplanting flooding) and confounding (saline storm-surge) pathways into the same SAR backscatter trough; the Module 02 saline-flood classifier intercepts the confounding pathway.

Figure 2. fig2_did_coefplot.png

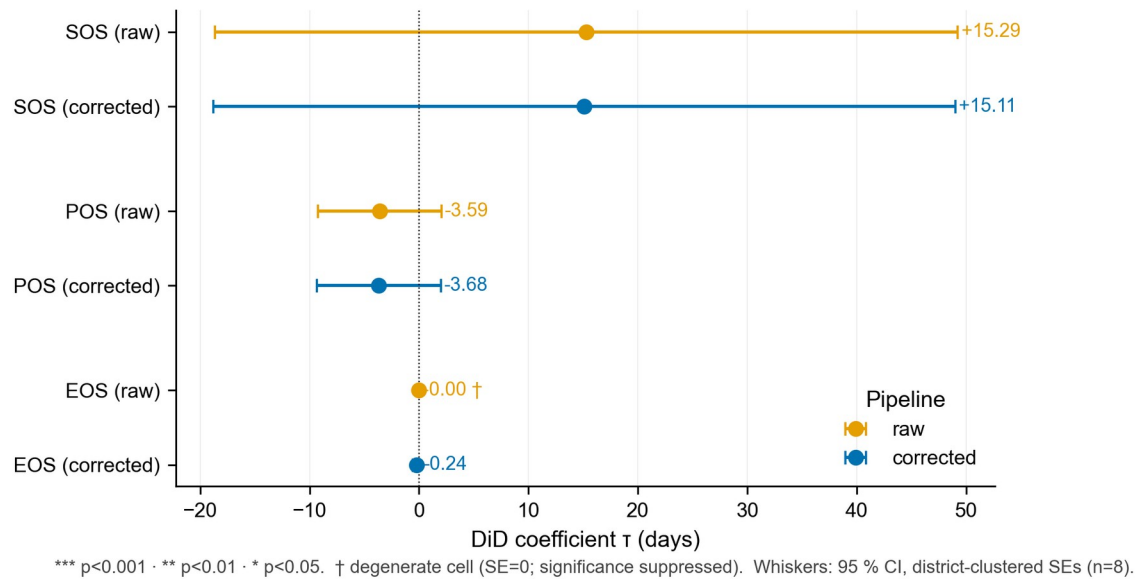
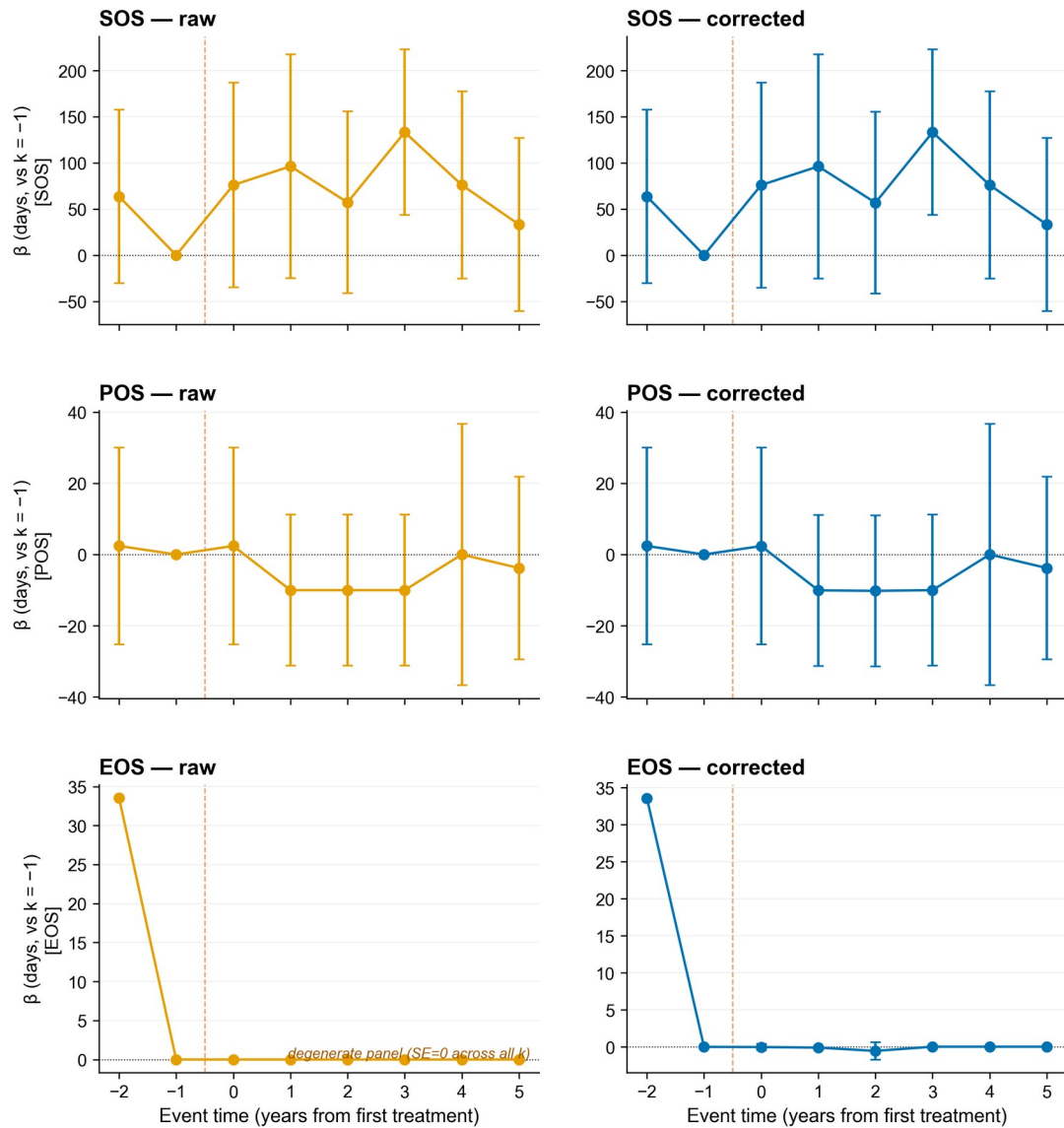


Figure 2. TWFE-DiD coefficients (raw vs. classifier-corrected panel). Point estimates with district-clustered (CR1) standard-error bars and 95% wild-cluster restricted bootstrap intervals for SOS, POS, and EOS, raw and corrected pipelines on the real v2.1 panel.

Figure 3. fig3_event_study.png



Reference period $k = -1$ (year before first treatment landfall, 2018). Whiskers: 95 % CI, district-clustered SEs.

Figure 3. Event-study plot, raw vs. corrected pipelines (Kharif 2017-2024). Coefficients at relative-year leads ($k = -2, -1$) test for pre-trends; coefficients at lags ($k = 0, 1, 2$) trace dynamic treatment effects; $k = -1$ (2018) is the omitted reference.

Figure 4. fig4_district_sos_panel.png

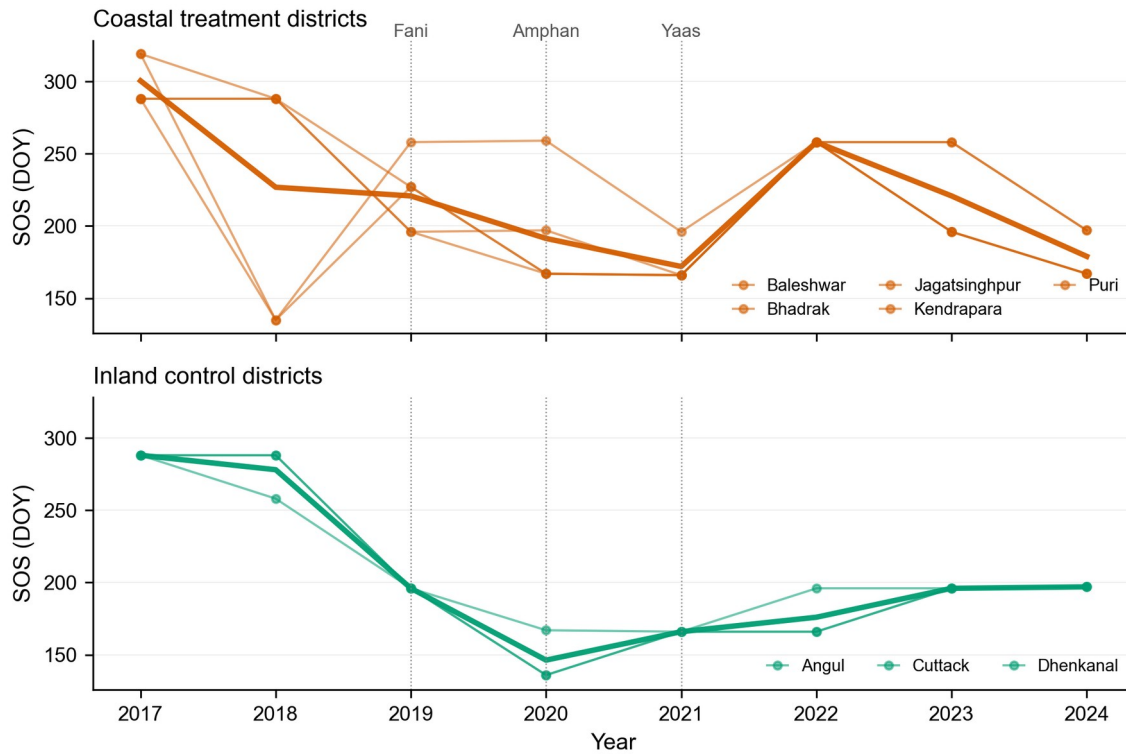


Figure 4. District-level SOS trajectories, raw vs. corrected (2017-2024). Solid lines show raw-pipeline SOS dates; dashed lines show classifier-corrected SOS dates; vertical bands mark cyclone landfall years. Treated coastal districts are coloured; inland controls are grey.

Figure 5. fig5_power_curves.png

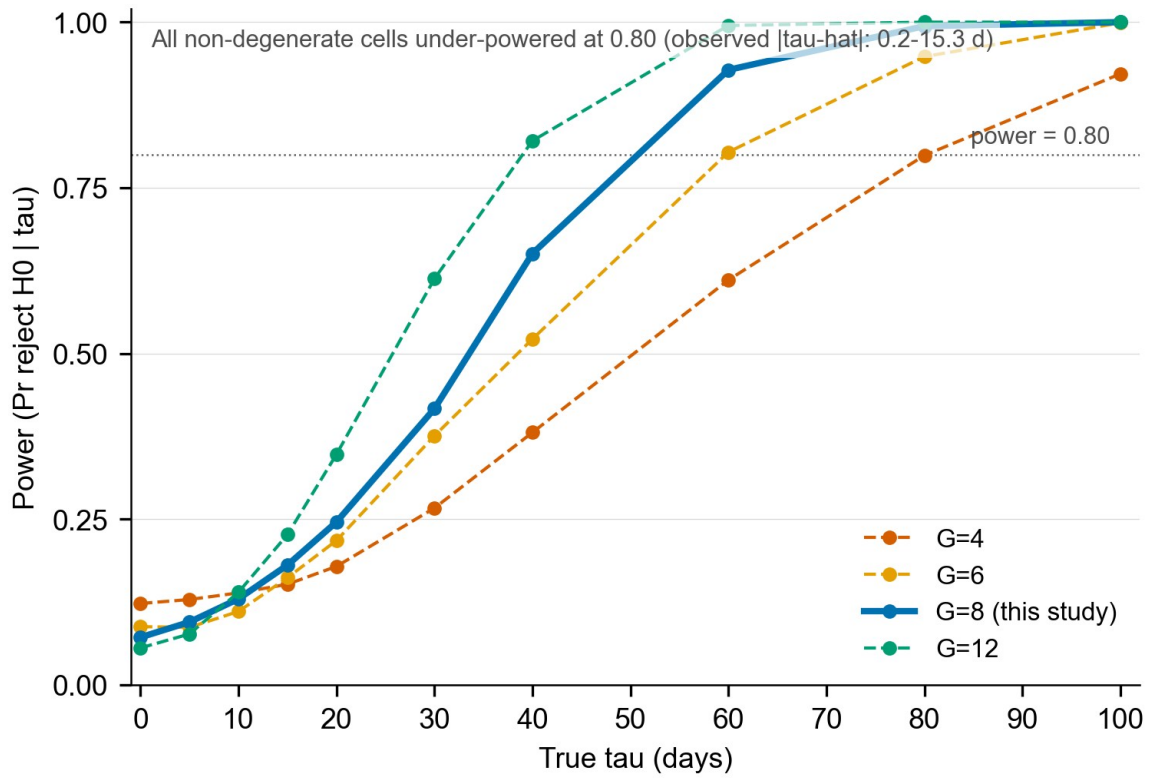


Figure 5. Empirical power curves from the Monte-Carlo simulation (Note S4). Rejection rate of H_0 against true effect size τ over 999 replications per grid point at $G = \{4, 6, 8, 12\}$ with empirical variance components $\sigma_u = 13.06$ d, $\sigma_t = 41.75$ d, $\sigma_{\epsilon} = 31.28$ d estimated from the real v2.1 corrected-SOS panel. At $G = 8$ (this study) power ≥ 0.80 is reached only for $\tau \geq 60$ d; the type-I rate under H_0 is 0.07, close to nominal 0.05.

Figure 6. fig6_placebo_distribution.png

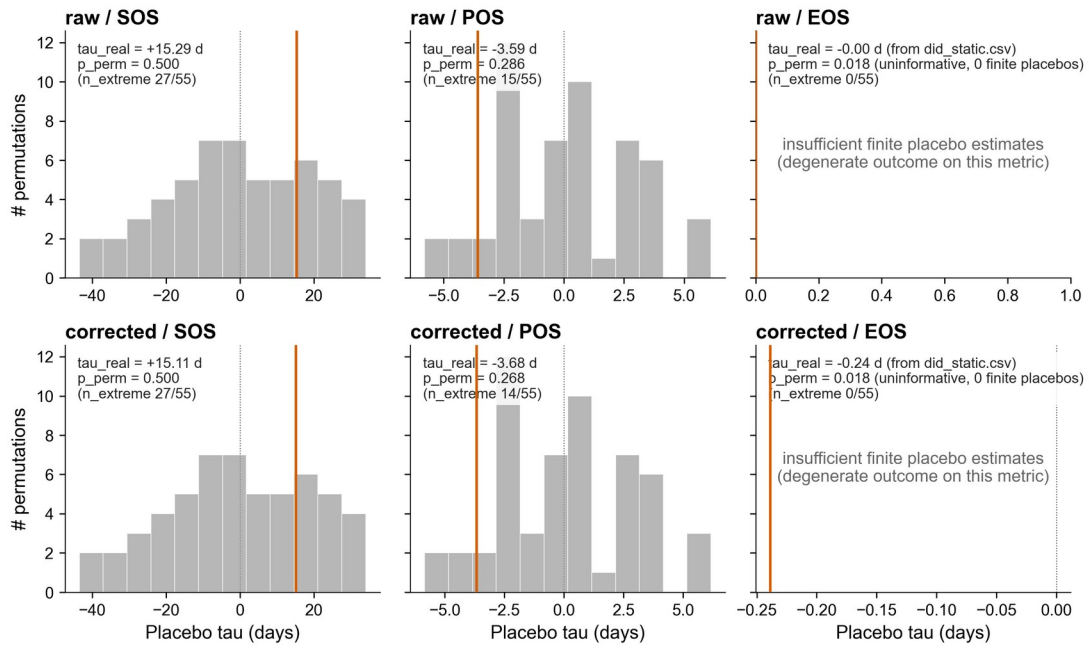


Figure 6. In-space placebo distributions (donor-swap, 56 permutations). Histograms of placebo τ_{hat} values from all swaps of treated/control labels across district pairs, with the observed τ_{hat} marked as a vertical line. The corrected/EOS cell yields $p_{\text{perm}} = 0.018$ (non-significant after Bonferroni correction across the six-cell family).

Figure S1. figS1_cyclone_climatology.png

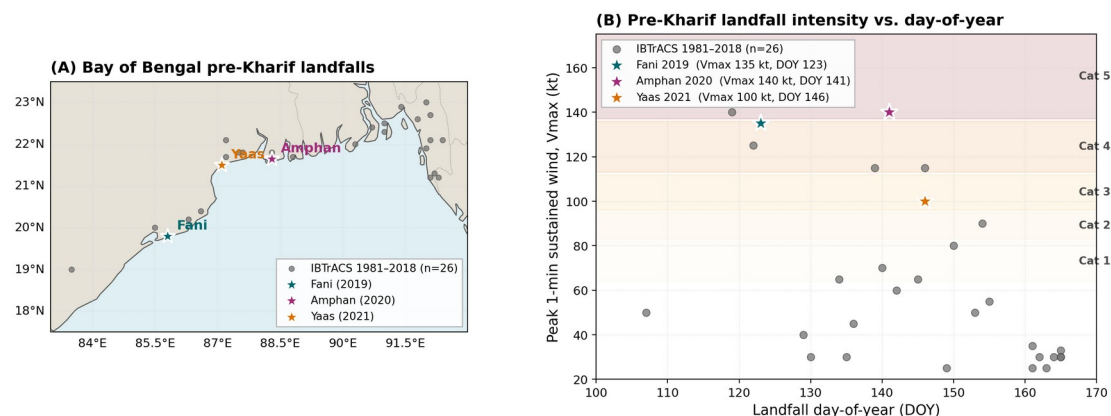


Figure S1. Cyclone climatology for the Bay of Bengal, 1980-2024. Annual landfall counts in the 50-km IBTrACS buffer around the coastal Odisha districts, stratified by Saffir-Simpson category and pre-monsoon vs. post-monsoon timing.

Figure S2. figS2_backscatter_signatures.png

Figure S2. Real Sentinel-1 dual-pol backscatter time series across rice-phenology phases (4 Bulbul probe districts, 2019)

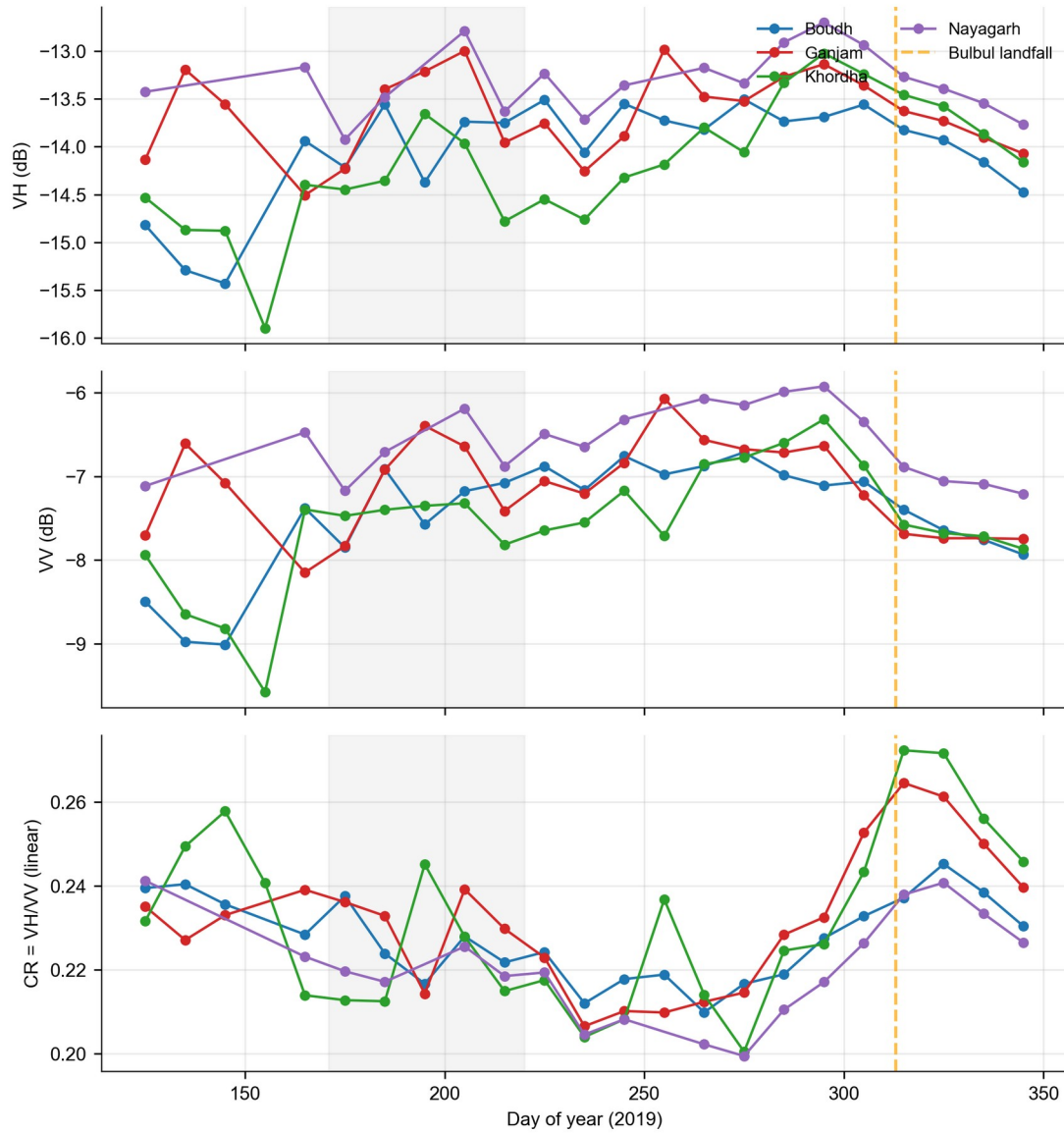


Figure S2. Sentinel-1 RTC dual-polarisation VH/VV/CR backscatter signatures across four phenological phases (pre-canopy, early canopy, peak canopy, event window) for the four Bulbul out-of-sample probe districts (Boudh, Ganjam, Khordha, Nayagarh), retrieved from Microsoft Planetary Computer.